

INTELLIGENT FRAMEWORK FOR DETECTING PREDATORY PUBLISHING VENUES

AN INDUSTRIAL ORIENTED MINI PROJECT REPORT

Submitted

In partial fulfilment of the requirement for the award of the degree of

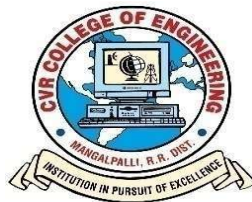
BACHELOR OF TECHNOLOGY
In
COMPUTER SCIENCE AND ENGINEERING

by

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2025-26



CVR COLLEGE OF ENGINEERING

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CERTIFICATE

This is to certify that the project entitled “Intelligent Framework for Detecting Predatory Publishing Venues” that is being submitted by Aithagoni Spandana (23B81A05B0), D. Sricharanpranav (23B81A05B3), in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering to the CVR College of Engineering, is a record of Bonafide work carried out by them under my guidance and supervision during the year 2025-2026.

The result embodied in this Industrial Oriented Mini Project work has not been submitted to any other University or Institute for the award of any degree or diploma.

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ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without the mention of the people who made it possible and whose encouragement and guidance has been a source of inspiration throughout the course of the project.

It is a great pleasure to convey our profound sense of gratitude to our Principal, **Dr. K. Ramamohan Reddy**, Vice-Principal, **Prof. L. C. Siva Reddy**, Head of CSE Department, **Dr. A. Vani Vathsala**, CVR College of Engineering, for having been kind enough to arrange necessary facilities for executing the project in the college.

We deem it a pleasure to acknowledge our sense of gratitude to our project guide **Dr. S. Srinivasulu** under whom we have carried out the project work. Her incisive and objective guidance and timely advice encouraged us with constant flow of energy to continue the work.

We wish a deep sense of gratitude and heartfelt thanks to management for providing excellent lab facilities and tools.

Finally, we thank all those whose guidance helped us in this regard. We place highest regards to our Parents, our Friends and Well-wishers who helped a lot in making the report of this project.

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ABSTRACT

This project is developed with the aim of helping researchers identify predatory journals and conferences using an intelligent and automated system. In today's research environment, many publishing venues are misleading and do not follow proper peer review or ethical standards, making it difficult for researchers to identify genuine platforms.

To address this issue, the proposed system analyses important details such as journal website content, indexing claims, peer review policies, publication fees, and editorial information. Using Machine Learning and Natural Language Processing (NLP), the system learns patterns from past data to detect suspicious publishing practices. The data is stored and managed using a SQLite database for efficient processing.

The system classifies publishing venues as Predatory or Legitimate and provides a simple interface for users to get quick and reliable results. It helps researchers make safe publishing decisions and maintain academic integrity.

In the future, the system can be enhanced with advanced models, multilingual support, and real-time verification features to improve accuracy and usability.

Keywords:

Predatory Journals, Academic Publishing, Machine Learning, Natural Language Processing (NLP), Research Integrity, Journal Classification, Peer Review Analysis, SQLite Database, Automated Detection System.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers
TF-IDF	Term Frequency–Inverse Document Frequency
UI	User Interface
API	Application Programming Interface
DB	Database
SQL	Structured Query Language
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
URL	Uniform Resource Locator
CNN	Convolutional Neural Network

1 INTRODUCTION

Predatory publishing venues have become a major problem in today's research environment. These journals and conferences attract researchers by offering quick publication, but they do not follow proper peer review or ethical standards. As a result, many researchers unknowingly publish in fake journals, which affects their academic credibility and research quality.

A predatory journal or conference is a publishing platform that prioritizes profit over quality and does not maintain proper publishing standards. These venues often provide false information about indexing, editorial boards and peer review processes, making it difficult for researchers to identify whether a publishing platform is genuine or fake.

Machine Learning (ML): Machine Learning is a technology that allows computers to learn patterns from data and make predictions without being explicitly programmed. In this project, ML is used to classify publishing venues as Predatory or Legitimate based on given features. Various classification algorithms such as logistic regression, decision trees or ensemble methods can be applied to improve prediction accuracy and model performance.

Natural Language Processing (NLP): NLP is a field of artificial intelligence that helps computers understand and analyse human language. In this system, NLP is used to analyse journal website content and extract meaningful information. Techniques such as text preprocessing, tokenization, stop-word removal and feature extraction are used to identify suspicious patterns in textual data.

BERT Model: BERT (Bidirectional Encoder Representations from Transformers) is an advanced deep learning model used for understanding the context of text. It helps improve accuracy in detecting whether a journal is predatory or legitimate. By analysing the contextual relationships between words in both directions, BERT enables deeper semantic understanding and enhances classification performance compared to traditional NLP models.

SQLite Database: SQLite is a lightweight database used to store and manage data efficiently. In this project, it is used to store extracted journal information and prediction results.

This project proposes an intelligent system that automatically analyses journal and conference details such as website content, indexing claims, peer review policies, publication fees and editorial information. Based on these features, the system predicts whether a publishing venue is Predatory or Legitimate. The system provides a simple and user-friendly interface that helps researchers make safe publishing decisions. It reduces manual effort, improves accuracy and supports maintaining research integrity.

1.1 MOTIVATION

In today's research environment, the number of predatory journals and conferences is increasing rapidly. These platforms mislead researchers by offering quick publication without proper peer review or ethical standards. Many researchers, especially students and beginners, find it difficult to identify genuine publishing venues due to lack of awareness and reliable verification methods.

This creates a need for an intelligent system that can help researchers verify the authenticity of journals easily and efficiently. The motivation behind this project is to develop a system that reduces manual effort, improves decision-making and protects researchers from publishing in fake journals. By providing quick and reliable results, the system aims to support academic integrity and ensure that research work is published in credible platforms.

1.2 PROBLEM STATEMENT

Many researchers face difficulty in identifying whether a journal or conference is genuine or predatory. Existing methods rely on manual checking of journal details such as indexing, peer review policies and editorial information, which is time-consuming and often unreliable. It also requires prior knowledge and awareness, which not all researchers possess. There is no centralized system that automatically analyses these features and provides accurate results. Researchers often depend on multiple sources or personal judgment, which increases the chances of mistakes. Moreover, some predatory journals present misleading information that appears genuine, making detection even more difficult. As a result, researchers may publish in predatory venues, leading to loss of time, money and academic credibility. This highlights the need for a reliable and automated system to assist researchers in making better publishing decisions.

1.3 PROJECT OBJECTIVES

The main objective of this project is to develop an intelligent system to detect predatory journals and conferences. The system aims to automatically collect and analyse journal and conference data, including important features such as indexing claims, peer review policies, publication fees and editorial information. It uses Machine Learning and Natural Language Processing (NLP) techniques to classify publishing venues as Predatory or Legitimate. The project also focuses on providing a simple and user-friendly interface that helps researchers easily verify journals and make safe publishing decisions. Additionally, the system is designed to be reliable, accurate and scalable for handling large amounts of data.

1.4 PROJECT REPORT ORGANIZATION

The project report is organized in a systematic and structured manner to provide a clear understanding of the proposed system for detecting predatory publishing venues.

It begins with the Cover Page, which includes the project title, student details, institution name and academic year. This is followed by the Certificate and Declaration, which confirm the authenticity and originality of the work.

The Acknowledgement section expresses gratitude to the guide, faculty and others who supported the project. The Abstract gives a summary of the project, including its objective, methodology and expected outcomes. The Table of Contents provides an organized list of chapters and sections for easy navigation.

Chapter 1: Introduction

This chapter introduces the concept of predatory publishing and explains the need for an automated detection system. It includes the motivation behind the project, the problem statement and the objectives. It also provides a brief overview of how the report is structured.

Chapter 2: Literature Survey

This chapter reviews existing research and methods used for identifying predatory journals. It discusses different approaches such as rule-based methods and machine learning techniques, along with their advantages and limitations. This helps in understanding the need for the proposed system.

Chapter 3: Requirement Analysis

This chapter describes the requirements needed to develop the system. It includes software requirements, hardware requirements and user requirements. It also explains the system specifications and constraints required for proper implementation.

Chapter 4: System Design

This chapter presents the design of the proposed system. It includes the system architecture and detailed explanations of diagrams such as use case, class, sequence, activity and object diagrams. It also explains the methods, algorithms, datasets and technology stack used in the system.

Chapter 5: Implementation and Testing

This chapter explain how the system is implemented using various technologies. It includes details about the development environment, system functionality and working process. It also presents the results obtained from the system along with testing and validation to ensure accuracy and performance.

Chapter 6: Conclusion and Future Scope

This chapter summarizes the overall work and highlights the achievements of the project. It also discusses possible future improvements, such as enhancing model accuracy, adding new features and expanding the system for broader use.

Chapter 7: References

This chapter lists all the books, research papers and online resources referred to during the development of the project.

2 LITERATURE SURVEY

2.1 EXISTING WORK

Beall J. et al. [1] is one of the earliest approaches used to identify predatory journals. It follows a rule-based method by checking indicators such as lack of transparency, false indexing claims, poor editorial practices and unethical publishing behaviour. It provides a structured checklist for evaluating journals and helps researchers identify suspicious patterns but relies on manual analysis and expert judgment.

Relevance: Forms the foundation for detecting predatory journals but is not automated or scalable.

Shamseer L. et al. [2] conducted a comparative study between predatory and legitimate journals, highlighting differences in peer review processes, editorial board credibility, indexing status and website quality. The study also identifies warning signs such as incomplete contact details, poor website design and unrealistic publication timelines, which are common in predatory platforms.

Relevance: Provides useful features and insights for detection but does not include an automated classification system.

Dadkhah M. et al. [3] proposed a Machine Learning-based approach that uses features extracted from journal websites to classify journals as predatory or legitimate. The model analyses textual and structural data, improving efficiency compared to manual methods. However, its performance depends on feature selection, and it lacks advanced NLP techniques for deeper context understanding.

Relevance: Introduces automation using Machine Learning and improves efficiency but has limitations in accuracy and feature representation.

Devlin J. et al. [4] introduced BERT, a deep learning model for Natural Language Processing that understands the contextual meaning of text by analysing words in both directions. It is highly effective for text classification and improves accuracy in detecting predatory journals when applied to website content analysis.

Relevance: Enhances detection performance by providing advanced contextual text analysis.

2.2 LIMITATIONS OF EXISTING WORK

Existing approaches for detecting predatory publishing venues have several limitations that affect their effectiveness and reliability. Most of the existing methods are either manual or based on basic techniques, making them inefficient for large-scale usage.

Many systems rely on manual verification of journal details, which is time-consuming and requires expert knowledge. This makes the process slow and unsuitable for handling large volumes of data.

Additionally, most approaches consider only a limited set of features such as website content or indexing claims. They do not analyse all important factors like peer review policies, publication fees and editorial board details together, which reduces the accuracy of detection.

Traditional Machine Learning models lack advanced text understanding capabilities. Without proper contextual analysis, the prediction accuracy may be low, leading to incorrect classification of journals.

There is also no centralized system that provides complete analysis, classification and result display in a single platform. Researchers often depend on multiple sources, which increases confusion and effort.

Furthermore, many existing systems do not use advanced deep learning models like BERT, which are capable of better contextual understanding and improved accuracy.

Finally, manual and semi-automated approaches are not scalable, as they cannot efficiently handle many journals and conferences in real-world scenarios.

3 REQUIREMENTS ANALYSIS

Visual Studio Code is used as the development environment for building the application. The system is designed as a web-based application, where the user interface is developed using HTML and CSS to provide a clean and responsive layout. JavaScript is used to handle user interactions and manage data flow between the frontend and backend.

The backend is implemented using Python with the Flask framework, which manages request handling, data processing and communication with the machine learning model. Machine Learning and Natural Language Processing (NLP) techniques are used to analyse journal data and perform classification. A SQLite database is used to store journal details, processed data and prediction results efficiently.

3.1 SOFTWARE REQUIREMENTS

- Operating System: Windows 7 or above
- Programming Language: Python
- Front-End: HTML, CSS, JavaScript
- Framework: Flask
- Libraries/Tools: Scikit-learn, NLTK, BERT
- Database: SQLite
- Development Tool: Visual Studio Code

3.2 HARDWARE REQUIREMENTS

- Processor – Dual-core processor, 1 GHz or above
- RAM – Minimum 4 GB recommended
- Storage – Minimum 2 GB free space

3.3 USER REQUIREMENTS

The system provides role-based access for users and administrators to ensure secure and controlled usage. Users can enter journal or conference URLs for analysis, and the system automatically processes the input to classify the publishing venue as predatory or legitimate.

The platform offers a simple and user-friendly interface that allows easy navigation, enabling users to quickly access features such as URL submission and result viewing. Users can view detailed results along with confidence scores, helping them make informed decisions about publishing venues.

The system supports real-time processing, allowing users to receive quick predictions without delays. It also maintains a history of previously analysed journals, enabling users to revisit past results when needed.

For administrators, the system provides functionalities to manage datasets, monitor system performance and update or retrain the machine learning model to improve accuracy. The admin can also manage stored data and ensure the system runs efficiently.

The system ensures data accuracy and reliability by using Machine Learning and NLP techniques. It also provides secure handling of user input and stored data. Overall, the platform is designed to be efficient, reliable and accessible to users with minimal technical knowledge.

4 SYSTEM DESIGN

4.0 PROPOSED SYSTEM ARCHITECTURE

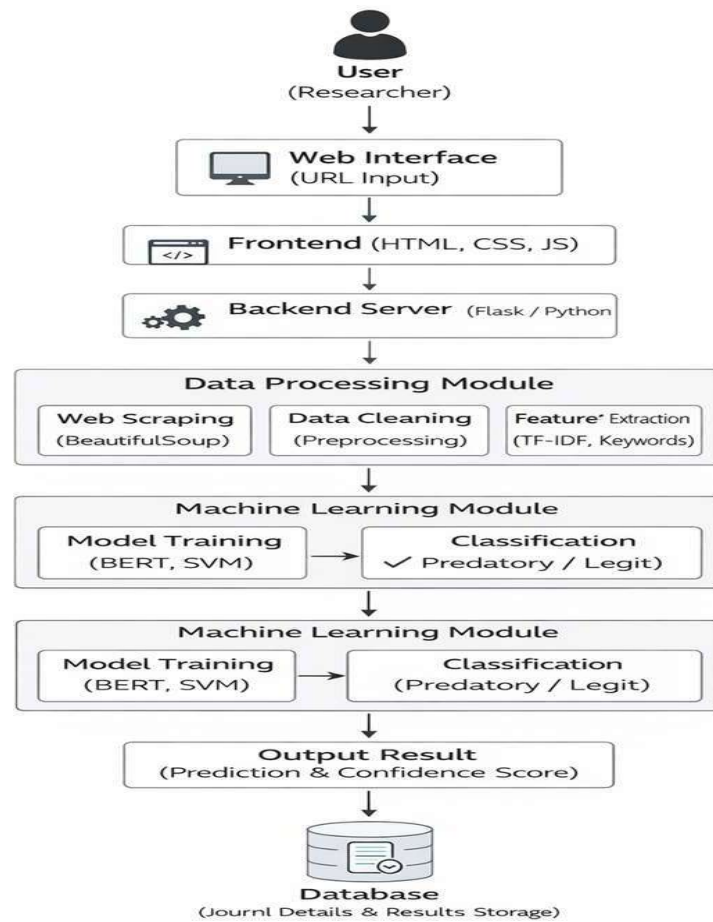


Fig:4.0.1 Proposed System Architecture

The Fig: 4.0.1 diagram titled “Proposed System Architecture” represents the workflow of the system designed to detect predatory publishing venues. The process begins with the user (researcher) entering a journal or conference URL through the web interface, which is sent to the backend server for processing. The system performs web scraping to extract relevant information from the journal website. The extracted data is then pre-processed by cleaning and preparing it for analysis. Feature extraction techniques such as TF-IDF are applied to convert the data into a structured format. The processed data is given to the machine learning model, which analyses the features and classifies the publishing venue as predatory or legitimate. The system then generates the result along with a confidence score and displays it to the user. Finally, all data and results are stored in the database for future reference. This architecture ensures an automated, accurate and efficient system for detecting predatory publishing venues.

4.1 PROPOSED METHODS / ALGORITHMS

The proposed system uses a structured approach combining Machine Learning and Natural Language Processing (NLP) to detect predatory publishing venues.

The process begins with the user providing a journal or conference URL as input. The system collects relevant data from the website using web scraping, including content, indexing claims, peer review policies, publication fees and editorial information.

The collected data is then pre-processed by cleaning, removing irrelevant content and standardizing the format. Feature extraction techniques such as TF-IDF are applied to convert the textual data into numerical features suitable for analysis.

These features are then used by classification models such as BERT, which analyse patterns in the data and identify whether the publishing venue is predatory or legitimate.

Finally, the system displays the prediction along with a confidence score to the user. The model can be periodically updated with new data to improve accuracy and overall performance.

4.2 CLASS/USE CASE/SEQUENCE/ACTIVITY DIAGRAM

- Use Case Diagram

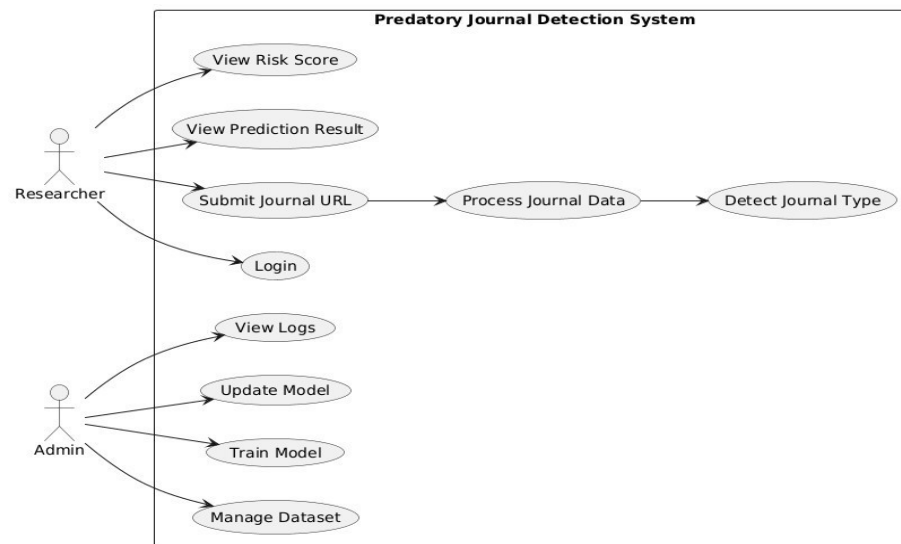


Fig: 4.2.1 Use Case Diagram

The Fig: 4.2.1 Use Case Diagram represents the interaction between users and the system for detecting predatory publishing venues. The main actors are the Researcher (User) and the Admin. The Researcher

interacts with the system by entering a journal or conference URL, submitting it for analysis and viewing the prediction results, which indicate whether the venue is predatory or legitimate along with a confidence score. The admin manages the system by updating datasets, monitoring performance and maintaining the machine learning model. The admin can also manage stored data and improve the model through retraining. This diagram clearly shows user interactions and ensures a simple, efficient and user-friendly process for identifying predatory publishing venues.

- **Class Diagram**

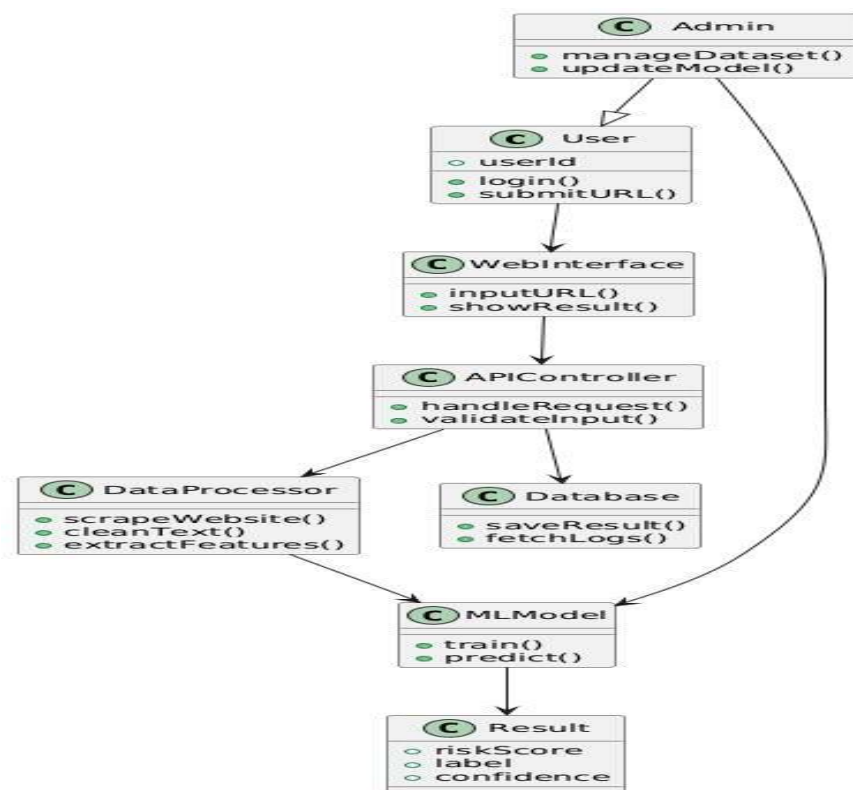


Fig: 4.2.2 Class Diagram

The Fig: 4.2.2 Class Diagram illustrates the structural design of the system used for detecting predatory publishing venues. It shows the main entities of the system, along with their functions and how they interact with each other. The User (Researcher) entity allows users to input journal or conference URLs and view prediction results. The admin entity is responsible for managing the system, including updating datasets, monitoring system performance and improving the machine learning model. The Journal Data entity stores important information such as journal URL, website content, indexing claims and peer review details, which

are used for analysis. The Data Processing Module performs operations such as web scraping, data cleaning and feature extraction to prepare the data for the model. The Machine Learning Model analyses the processed data and classifies the publishing venue as predatory or legitimate. The Result entity stores the prediction output along with a confidence score and the Database maintains all journal data and results for future reference. This diagram provides a clear understanding of the system's structure and ensures proper interaction between components, resulting in an efficient and scalable solution.

- **Sequence Diagram**

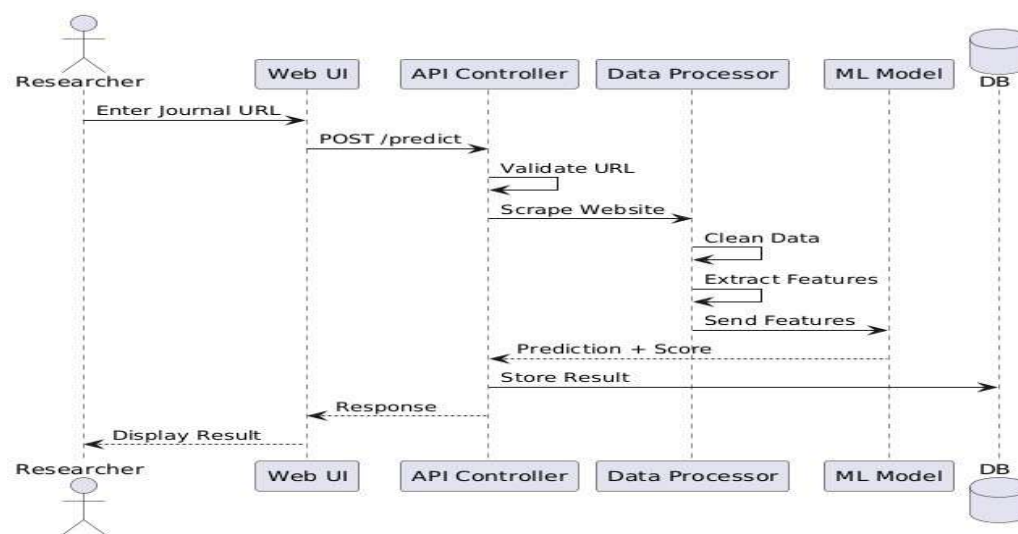


Fig: 4.2.3 Sequence Diagram

The Fig: 4.2.3 Sequence Diagram illustrates the interaction between the User (Researcher) and the system during the process of detecting predatory publishing venues. It shows the sequence of operations from input to result generation. The process begins when the user enters a journal or conference URL through the web interface. The system receives the input and sends it to the backend server for processing. The backend performs data extraction from the journal website using web scraping techniques, followed by preprocessing and feature extraction. The processed data is then passed to the machine learning model, which analyses the features and classifies the publishing venue as predatory or legitimate. The system then generates the result along with a confidence score and displays it to the user. Finally, the system stores the input data and prediction results in the database for future reference. This diagram highlights a smooth and structured interaction between the user and the system, ensuring an efficient and automated detection process.

- **Activity Diagram**

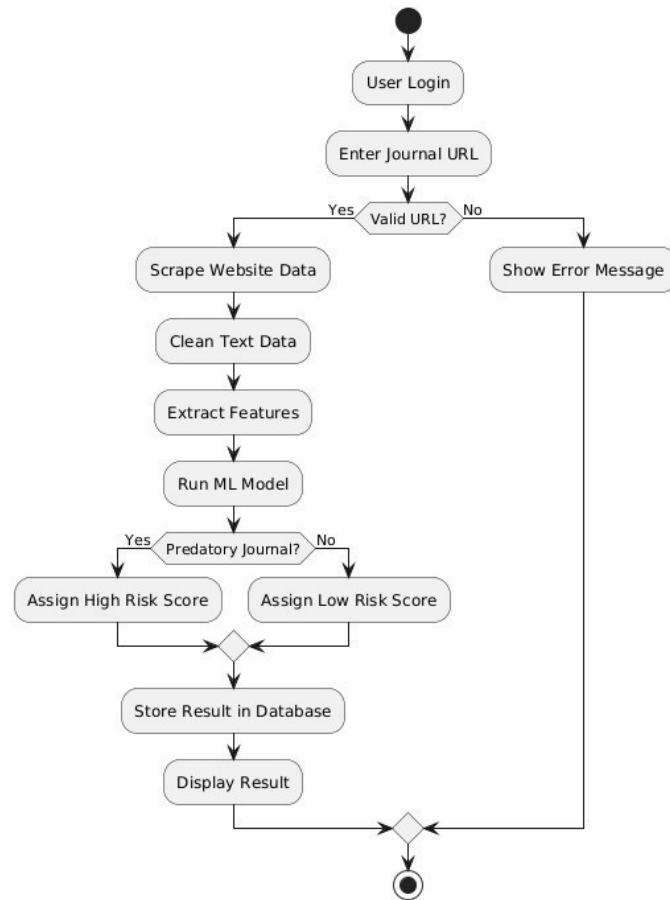


Fig: 4.2.4 Activity Diagram

The Fig: 4.2.4 Activity Diagram represents the workflow of the system used to detect predatory publishing venues, showing the process from user input to result generation. The process begins when the user enters a journal or conference URL. The system validates the input; if it is invalid, an error message is displayed and the process stops. If valid, the system performs data extraction using web scraping to collect relevant journal information. The extracted data is then pre-processed and cleaned, followed by feature extraction to convert it into a structured format. This data is passed to the machine learning model, which analyses the features and classifies the publishing venue as predatory or legitimate. Finally, the system displays the result along with a confidence score and stores the data in the database for future reference. This diagram provides a clear and logical representation of the workflow, ensuring an efficient and automated detection process.

- Object diagram

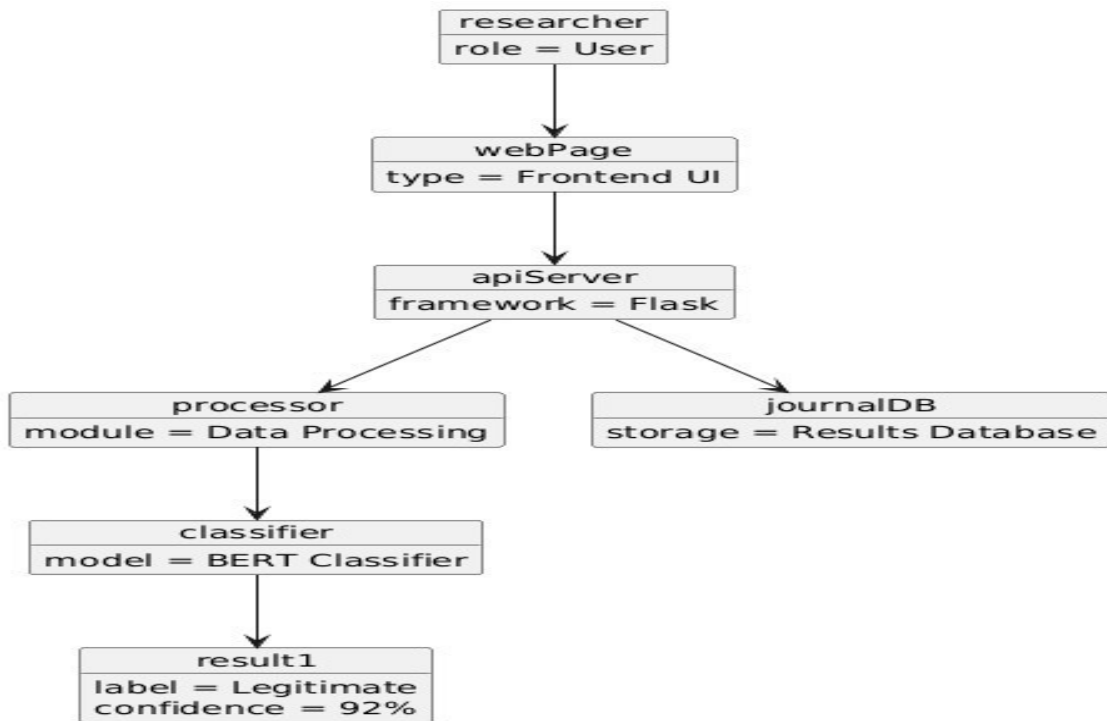


Fig: 4.2.5 Object Diagram

The Fig: 4.2.5 Object Diagram represents a snapshot of the system showing real-time instances of objects involved in detecting predatory publishing venues. It illustrates how different components of the system interact at a particular moment.

The diagram begins with the researcher who interacts with the web page to provide a journal or conference URL. This request is handled by the API server, which processes the input.

The backend communicates with the data processing module, where the input data is prepared for analysis, and interacts with the journal database, which stores journal details and results. The processed data is then passed to the classifier, which performs classification.

Finally, the system generates the output in the form of a result object that includes the label along with a confidence score. This diagram provides a clear representation of how system components exist and interact at runtime, ensuring an efficient and structured detection process.

4.3 DATASETS AND TECHNOLOGY STACK

The proposed system uses different types of datasets to accurately detect whether a journal or conference is predatory or legitimate.

The Journal Dataset includes details such as journal names, website URLs, indexing information, peer review policies, publication fees and editorial board details. This data is collected from publicly available sources and through web scraping, forming the core input for analysis.

The Website Content Data consists of textual information extracted from journal websites. This data is processed using Natural Language Processing (NLP) techniques to identify patterns, keywords and suspicious content associated with predatory publishing practices.

The Training Dataset contains labelled data where journals are classified as predatory or legitimate. This dataset is used to train machine learning models, enabling them to learn patterns and improve prediction accuracy over time.

The system also uses User Input Data, which includes the journal or conference URLs entered by users for analysis. This allows real-time evaluation of publishing venues.

Finally, all processed data and prediction results are stored in the database for future reference, analysis and model improvement.

Technology Stack

The system is developed using modern technologies to ensure efficiency, scalability and accurate detection of predatory publishing venues.

The frontend of the application is built using HTML, CSS and JavaScript, which provides a simple, responsive and user-friendly interface. It allows users to easily enter journal or conference URLs and view prediction results.

The backend is developed using Python with the Flask framework, which handles request processing, communication between modules and integration with the machine learning model. Flask is lightweight and suitable for building scalable web applications.

For data processing and analysis, Machine Learning and Natural Language Processing (NLP) techniques are used. Libraries such as Scikit-learn are used for implementing classification algorithms, while NLTK is used for text preprocessing tasks like tokenization and cleaning.

The BERT model is used for advanced contextual text analysis, improving the accuracy of classification.

The system uses BeautifulSoup for web scraping to extract relevant information from journal websites. Feature extraction techniques such as TF-IDF are applied to convert textual data into numerical form for model processing.

The database used is SQLite, which stores journal details, processed data and prediction results efficiently. It is lightweight and does not require a separate server, making it suitable for this application.

Development and coding are carried out using Visual Studio Code, which provides a flexible environment with support for Python and web technologies.

Overall, this technology stack ensures an efficient, reliable and scalable system for detecting predatory publishing venues.

5 IMPLEMENTATIONS

5.1 SYSTEM SCREENSHOT

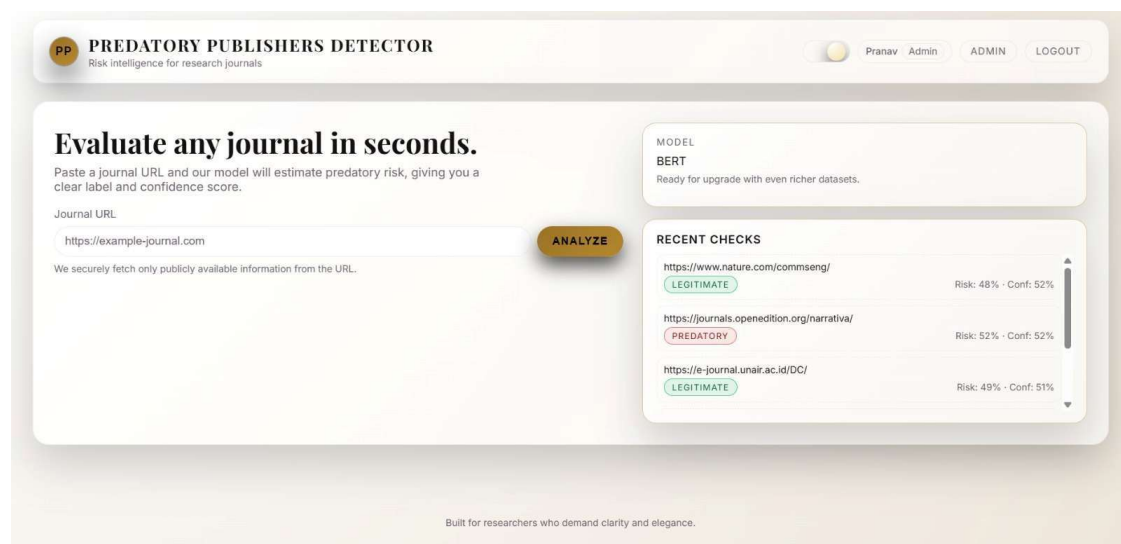


Fig:5.1.1 User Interface – Journal Analysis Page

The Fig: 5.1.1 shows the main user interface of the Predatory Publishers Detector system, which is designed to help researchers quickly evaluate journals or conferences by entering a URL. The interface is developed to be simple, intuitive and efficient, ensuring ease of use for users with minimal technical knowledge.

At the top of the page, the system displays the application title along with navigation options such as user profile, admin access and logout, allowing users to easily manage their session and access different functionalities.

The central section contains a URL input field, where users can paste the journal or conference link. Along with this, an “Analyse” button is provided to initiate the detection process. Once the user submits the URL, the system processes the input and begins analysis using backend modules and machine learning models.

On the right side of the interface, the system displays model information (BERT), indicating the type of model used for classification. Below this, the Recent Checks section shows a list of previously analysed journal URLs along with their classification results (predatory or legitimate) and corresponding confidence scores. This feature helps users track past analyses and compare results easily.

The interface follows a clean and organized design with proper alignment and spacing, which enhances readability and usability. Overall, this user interface enables fast, efficient and user-friendly interaction, allowing researchers to make informed publishing decisions with minimal effort.

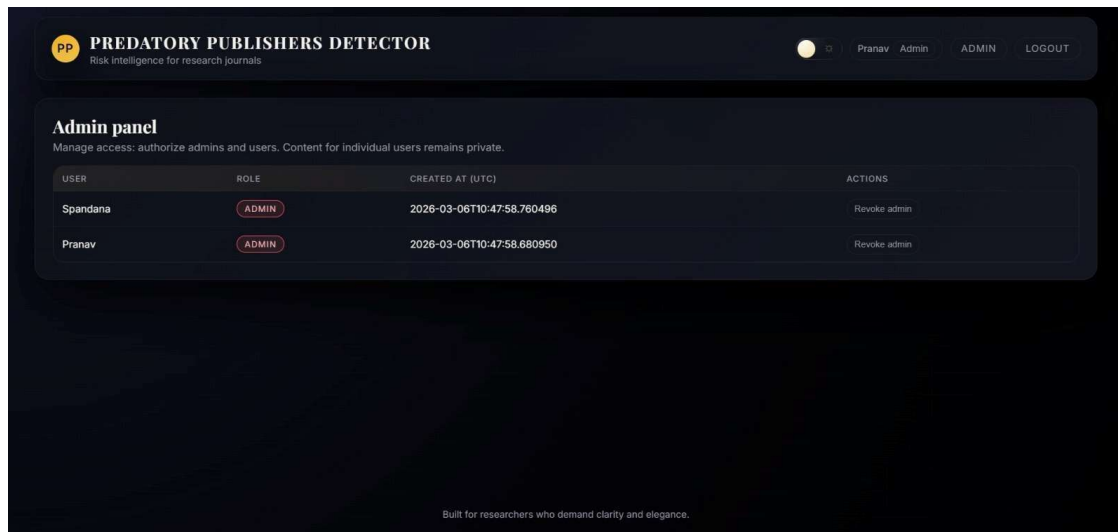


Fig: 5.1.2 Admin Panel Interface

The Fig: 5.1.2 illustrates the Admin Panel interface of the Predatory Publishers Detector system, which is designed to manage users and control access within the platform. This panel is accessible only to authorized administrators and plays a key role in maintaining system security and efficiency.

The interface displays a structured table containing user details such as username, role and account creation timestamp. Each user is assigned a specific role, such as admin, which determines their level of access and permissions within the system.

The panel also provides action controls, allowing administrators to manage user roles by granting or revoking admin privileges. This ensures that only authorized users have access to critical system functionalities such as dataset management and model updates.

Additionally, the admin panel helps in monitoring system usage and maintaining data integrity by providing a clear overview of registered users and their roles. It supports efficient administration by enabling quick updates and role management without affecting the overall system performance.

The interface is designed with a clean and organized layout, making it easy for administrators to navigate and perform tasks effectively. Overall, the admin panel enhances system security, user management and operational control.

The image shows a web interface for the 'Predatory Publishers Detector'. At the top, there is a header bar with a logo on the left, the title 'PREDATORY PUBLISHERS DETECTOR' in bold, and a subtitle 'Risk Intelligence for research journals'. On the right side of the header, there are three buttons: 'LOGIN', 'SIGN UP', and 'GUEST'. Below the header, the main content area is titled 'Create an account'. Under this title, a note states: 'Registered users get a personal history of checks. Guests stay anonymous.' There are two input fields: 'Username' and 'Password'. Below these fields is a prominent orange button labeled 'SIGN UP'. At the bottom of the page, there is a small line of text: 'Built for researchers who demand clarity and elegance.'

Fig: 5.1.3 User Registration Page

The Fig: 5.1.3 illustrates the User Registration (Sign Up) interface of the Predatory Publishers Detector system. This page is designed to allow new users to create an account and access personalized features of the system.

At the top of the interface, the application title is displayed along with navigation options such as Login, Sign Up and Guest access, allowing users to easily switch between different modes of usage. This provides flexibility for both registered and non-registered users.

The main section of the page contains a registration form, where users are required to enter details such as username and password. These credentials are used to create a secure account, enabling authenticated access to the system. Proper input validation is applied to ensure that the entered details meet required formats and security standards.

A “Sign Up” button is provided to submit the registration form. Once the user successfully registers, the account is created and the user can log in to access system functionalities. Registered users benefit from additional features such as maintaining a history of previously analysed journals, which helps in tracking and reviewing past results.

The interface also highlights that guest users can access the system anonymously, but without the ability to store or retrieve past data. This ensures both convenience and flexibility in usage.

The overall design is clean, minimal and user-friendly, with proper alignment and spacing, making it easy for users to complete the registration process without confusion. This page plays a crucial role in enabling secure access, personalized experience and efficient usage of the system.

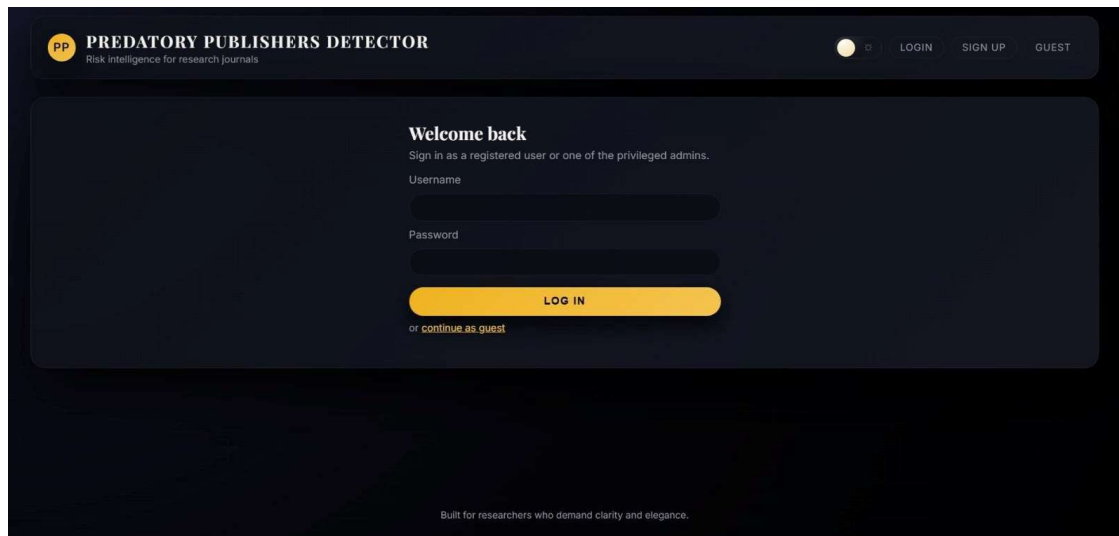


Fig: 5.1.4 User Login Page

The Fig: 5.1.4 illustrates the User Login interface of the Predatory Publishers Detector system, which allows registered users and administrators to securely access the platform.

At the top of the page, the system title is displayed along with navigation options such as Login, Sign Up and Guest access, providing users with multiple options to access the system based on their needs.

The central section contains a login form, where users are required to enter their username and password. These credentials are validated by the system to ensure secure authentication and authorized access. A “Log In” button is provided to submit the details and proceed to the main application interface.

Additionally, the interface includes an option to “continue as guest”, allowing users to access basic functionalities without creating an account. This provides flexibility for users who want quick access without storing their data.

The design of the page is clean, simple and user-friendly, with proper alignment and spacing, ensuring that users can easily enter their credentials and log in without confusion.

Overall, this login page plays a crucial role in maintaining system security, enabling controlled access and providing a smooth entry point to the application.

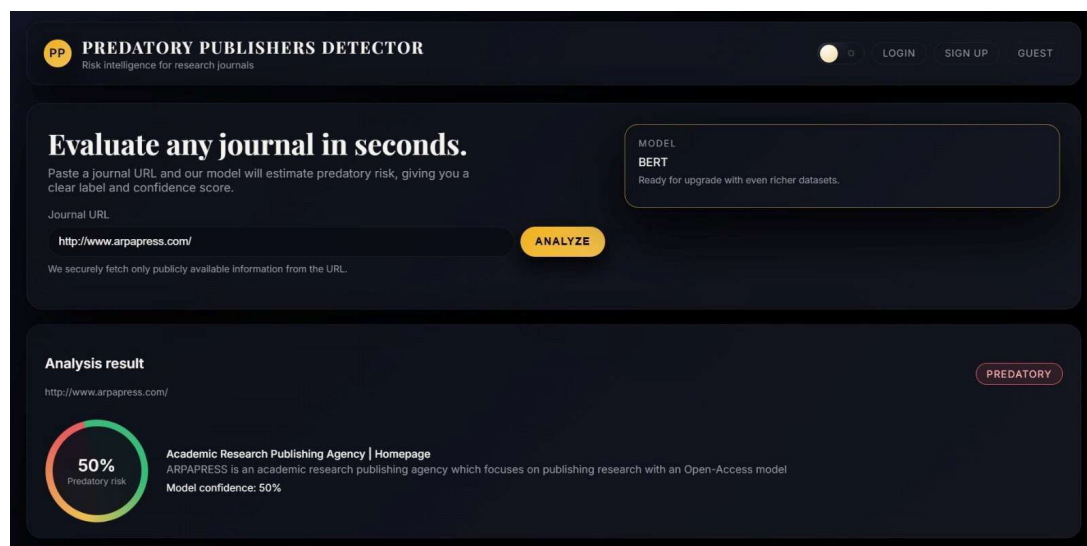


Fig: 5.1.5 Result Analysis Page

The Fig: 5.1.5 illustrates the Result Analysis Page of the Predatory Publishers Detector system, where the output of the journal evaluation process is displayed. This page provides users with detailed insights into whether a publishing venue is predatory or legitimate.

At the top, the interface displays the system title along with navigation options such as Login, Sign Up and Guest access, allowing users to manage their session. The main section contains a URL input field along with an “Analyse” button, enabling users to submit a journal link for evaluation.

Once the analysis is performed, the system displays the results in a structured format. The Analysis Result section shows the evaluated journal URL along with a visual representation of the predatory risk percentage using a circular indicator. This graphical representation helps users quickly understand the level of risk associated with the journal.

Along with the visual indicator, the system provides detailed information about the journal, including a brief description and the model confidence score, which indicates the reliability of the prediction. The result is clearly labelled as “Predatory” or “Legitimate”, making it easy for users to interpret the outcome.

Additionally, the interface displays the model used (BERT), indicating that advanced Natural Language Processing techniques are applied for accurate classification. The clean layout and organized presentation of results ensure that users can quickly understand and interpret the analysis without confusion.

Overall, this page plays a crucial role in delivering clear, accurate and user-friendly results, enabling researchers to make informed decisions about publishing venues.

5.2 RESULTS AND DISCUSSION

Results

The proposed system effectively detects whether a journal or conference is predatory or legitimate by analysing key features such as website content, indexing claims, peer review policies, publication fees and editorial details. The use of Machine Learning and Natural Language Processing (NLP) techniques improves the accuracy and reliability of classification compared to manual methods.

The system provides quick results by allowing users to enter a journal URL and receive a prediction along with a confidence score. The user-friendly interface ensures that researchers can easily use the system without requiring technical knowledge. The use of advanced models such as BERT enhances the system's ability to understand textual data and improves detection performance.

The system also stores results in the database, enabling efficient data management and future reference. Overall, the system reduces manual effort and supports researchers in making safe and informed publishing decisions.

Discussion

This project demonstrates the effective application of Machine Learning and NLP in solving real-world problems related to academic publishing. The system provides an automated and centralized solution for identifying predatory publishing venues, thereby improving research quality and reducing risks.

However, certain challenges exist. The accuracy of the system depends on the quality and size of the training dataset, and incomplete or outdated data may affect predictions. Additionally, some journal websites may restrict data access or frequently update their content, which can impact data extraction.

Continuous updates to the dataset and model are required to maintain accuracy and reliability. Despite these challenges, the system offers a scalable and efficient solution that can be further enhanced with advanced models and larger datasets.

5.3 TESTING

Test Scenario 1: URL Submission

Test Case ID: URL-1A

Test Case Description: User submits journal/conference URL

Test Priority: High

Prerequisite: User is on homepage

Post Requisite: URL is accepted for processing

S.No	Action	Inputs	Expected Output	Actual Output	Test Browser	Test Result	Test Comments
1	Enter journal URL	Valid URL	URL accepted and processing starts.	URL accepted and processing starts.	Chrome	PASS	NA

Tab:5.3.1 URL-1A-Journal URL Submission

This test case verifies the functionality of submitting a journal or conference URL for analysis in the system. The user enters a valid URL in the input field provided on the homepage and initiates the process.

The system checks whether the entered URL is in the correct format and accepts it for further processing. Upon successful submission, the system begins the analysis by sending the input to the backend for data extraction and processing.

The expected outcome is that the system accepts the valid URL and starts the detection process without any errors. The actual result confirms that the system behaves as expected, indicating successful functionality of the URL submission feature.

Test Scenario 2: Invalid URL Handling

Test Case ID: URL-2A

Test Case Description: System handles invalid input

Test Priority: High

Prerequisite: None

Post Requisite: Error message displayed

S.No	Action	Inputs	Expected Output	Actual Output	Test Browser	Test Result	Test Comments
1	Enter invalid URL	Random text	Error message displayed.	Error message displayed	Chrome	PASS	NA

Tab: 5.3.2 URL-2A – Invalid Input Handling

This test case verifies the system's ability to handle invalid input during URL submission. The user enters incorrect or random text instead of a valid journal or conference URL in the input field.

The system checks the format of the input and identifies it as invalid. Instead of proceeding with processing, the system displays an appropriate error message to inform the user about the incorrect input.

The expected outcome is that the system should reject invalid inputs and prevent further processing. The actual result confirms that the system correctly displays an error message, ensuring proper input validation and improving system reliability.

Test Scenario 3: Data Extraction

Test Case ID: Data-1A

Test Case Description: Extract journal website data

Test Priority: High

Prerequisite: Valid URL entered

Post Requisite: Data extracted successfully

S.No	Action	Inputs	Expected Output	Actual Output	Test Browser	Test Result	Test Comments
1	Process URL	Valid journal URL	Website data extracted	Website data extracted	Chrome	PASS	NA

Tab: 5.3.3 Data-1A – Data Extraction

This test case verifies the system's ability to extract relevant information from the journal website after a valid URL is submitted. Once the user provides a valid URL, the system initiates the data extraction process using web scraping techniques.

The system retrieves important details such as website content, indexing claims, peer review policies and other relevant information required for analysis.

The expected outcome is that the system successfully extracts the necessary data from the given URL without errors. The actual result confirms that the system accurately extracts the required data, ensuring that the input is properly prepared for further processing and classification.

Test Scenario 4: Prediction Generation

Test Case ID: ML-1A

Test Case Description: Classification of journal

Test Priority: High

Prerequisite: Data processed

Post Requisite: Prediction generated

S.No	Action	Inputs	Expected Output	Actual Output	Test Browser	Test Result	Test Comments
1	Run model	Processed data	Predatory/Legitimate result	Predatory/Legitimate result	Chrome	PASS	NA

Tab: 5.3.4 ML-1A – Prediction Generation

This test case verifies the system's ability to generate predictions using the machine learning model. After the data is successfully processed and prepared, it is passed to the classification model for analysis.

The model analyses the extracted features and applies Machine Learning and NLP techniques to determine whether the journal or conference is predatory or legitimate.

The expected outcome is that the system generates an accurate classification result based on the processed data. The actual result confirms that the system correctly produces the prediction, ensuring proper functioning of the machine learning module.

Test Scenario 5: Result Display

Test Case ID: Result-1A

Test Case Description: Display prediction to user

Test Priority: High

Prerequisite: Prediction completed

Post Requisite: Result visible to user

S.No	Action	Inputs	Expected Output	Actual Output	Test Browser	Test Result	Test Comments
1	View result	Prediction output	Result with confidence score displayed	Result displayed	Chrome	PASS	NA

Tab: 5.3.5 Result-1A – Result Display

This test case verifies the system's ability to display the prediction result to the user after the analysis is completed. Once the machine learning model generates the classification, the system presents the result on the user interface.

The result includes the classification label (predatory or legitimate) along with a confidence score, helping users understand the reliability of the prediction. The information is displayed in a clear and user-friendly format.

The expected outcome is that the result is displayed correctly without any delay or error. The actual result confirms that the system successfully shows the prediction output, ensuring proper interaction between the backend processing and frontend display.

Test Scenario 6: Database Storage

Test Case ID: DB-1A

Test Case Description: Store results in database

Test Priority: Medium

Prerequisite: Prediction completed

Post Requisite: Data stored successfully

S.No	Action	Inputs	Expected Output	Actual Output	Test Browser	Test Result	Test Comments
1	Save data	Journal data and result	Data stored in database	Data stored	Chrome	PASS	NA

Tab: 5.3.6 DB-1A – Data Storage

This test case verifies the system's ability to store journal data and prediction results in the database. After generating the prediction, the system saves the input URL along with the classification result and related details.

The database ensures proper storage and retrieval of data for future reference, allowing users or administrators to access previous records when needed.

The expected outcome is that the data is stored accurately without any loss or duplication. The actual result confirms that the system successfully stores the data, ensuring reliable data management and system consistency.

5.4 VALIDATION

Overview

Validation is the process of checking whether the system works correctly and meets all requirements. In this project, validation ensures that the system can accurately detect whether a journal is predatory or legitimate. It also checks that the system handles user input properly, processes data correctly and stores results safely. The goal is to make sure the system is reliable, secure and efficient for real-world use.

Validation Criteria

The system is validated based on several important criteria:

- **Input Validation:** The system should accept only valid journal or conference URLs. If the user enters invalid input, an appropriate error message must be displayed.
- **Data Accuracy:** The extracted journal information and prediction results should be correct and stored without errors or duplication.

- **Model Accuracy:** The machine learning model should correctly classify journals based on the given data. High accuracy is required to ensure reliable results.
- **Security:** User input and stored data must be protected from unauthorized access and misuse.
- **Performance:** The system should respond quickly and handle multiple requests without slowing down or crashing.

Validation Process

The validation process starts by checking the user input. The system verifies whether the entered URL is valid before processing. If the input is valid, the system proceeds to extract data from the journal website using web scraping techniques.

Next, the extracted data is cleaned and pre-processed to remove unnecessary content. Feature extraction is then performed to convert the data into a format suitable for the machine learning model.

The processed data is passed to the model, which analyses it and generates a prediction (predatory or legitimate). The system then checks whether the result is correctly displayed along with a confidence score.

Finally, the system ensures that all data and results are stored properly in the database. Performance testing is also carried out to check whether the system works efficiently under different conditions and workloads.

Validation Techniques Used

- **Unit Testing:** Each component of the system, such as data extraction, preprocessing and prediction, is tested individually to ensure correct functionality.
- **Integration Testing:** Different modules like frontend, backend, database and machine learning model are tested together to ensure smooth interaction.
- **User Acceptance Testing (UAT):** The system is tested from the user's perspective to ensure it is easy to use and meets user requirements.
- **Security Testing:** The system is checked for vulnerabilities to ensure that data is safe and secure.
- **Performance Testing:** The system is tested for speed, response time and ability to handle multiple users efficiently.

6 CONCLUSIONS

6.1 CONCLUSION

The proposed system for detecting predatory publishing venues has been successfully developed and tested to meet both functional and non-functional requirements. The system analyses journal data using Machine Learning and Natural Language Processing (NLP) techniques to accurately classify publishing venues as predatory or legitimate.

The use of an automated approach reduces manual effort, saves time and improves accuracy compared to traditional methods. The system also provides a simple and user-friendly interface, enabling researchers to easily verify journals and make informed publishing decisions without requiring technical expertise.

Additionally, the system ensures efficient data processing, reliable result generation and proper storage of information for future reference. Overall, the project offers a scalable, efficient and reliable solution that helps improve research quality and protects academic integrity.

6.2 FUTURE SCOPE

In the future, the system can be further enhanced by improving the accuracy of the machine learning models using larger and more diverse datasets. The integration of advanced deep learning techniques can further improve classification performance and reliability.

Additional features such as detection of fake conference invitations, email-based scam detection and multi-language support can be implemented to make the system more comprehensive and accessible to a wider audience.

The system can also be developed as a browser extension or mobile application to provide real-time journal verification. Deployment on cloud platforms can improve scalability and allow access to a global database of predatory journals, making the system more useful for researchers worldwide.

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